

Name: _____

Date: _____

Class: _____

Element D: Design Concept Generation, Analysis, and Selection

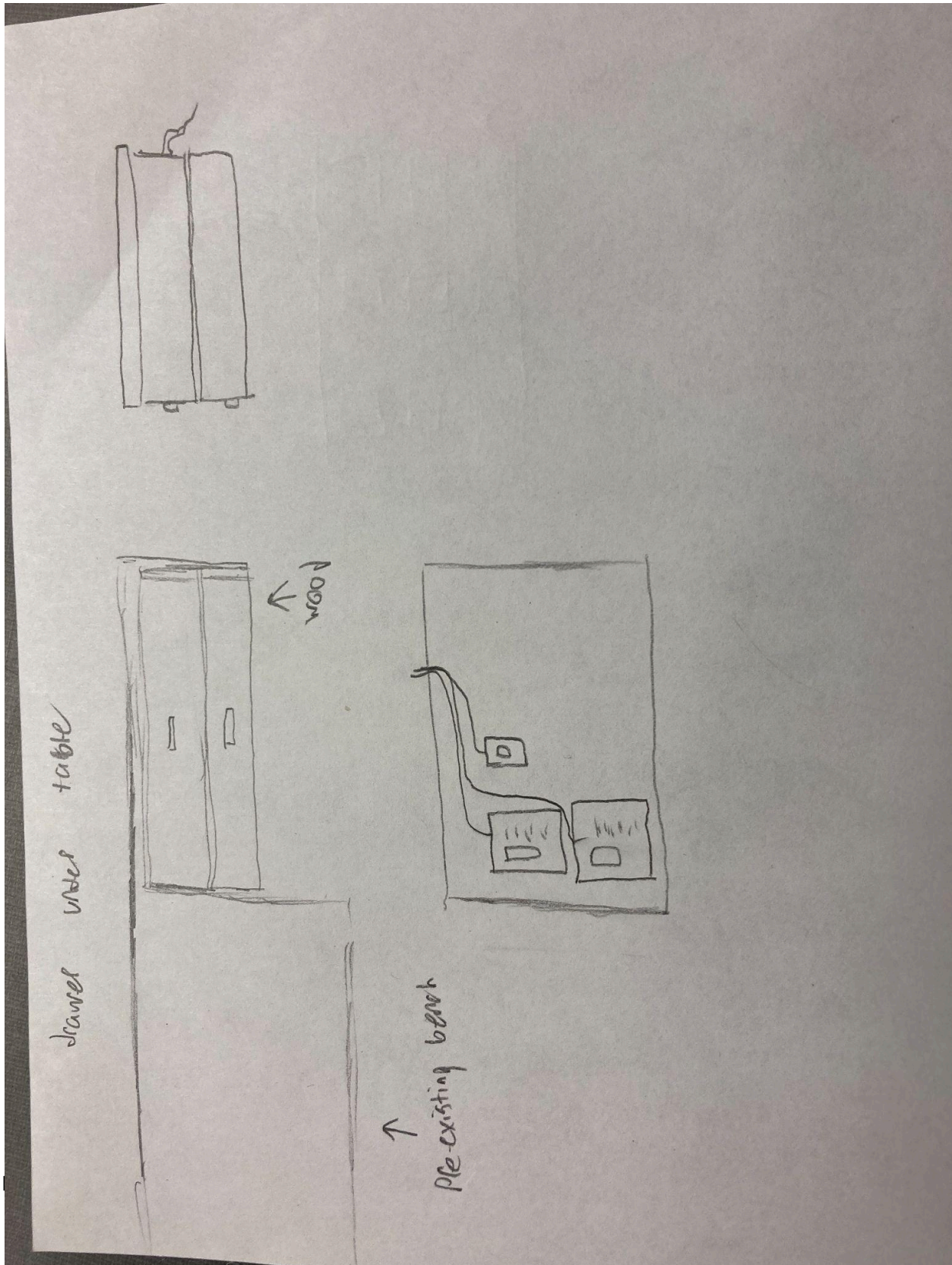
Design Idea Generation (D1)

This section focuses on the process of generating solutions. The initial rounds should include every idea you come up with but the subsequent rounds of ideation should lead to viable options. Each idea should include not only words, but also a picture. Use as many copies of this sub-template as

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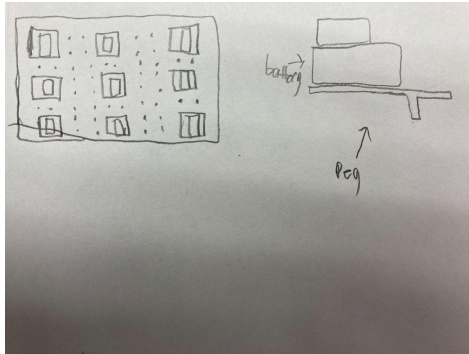
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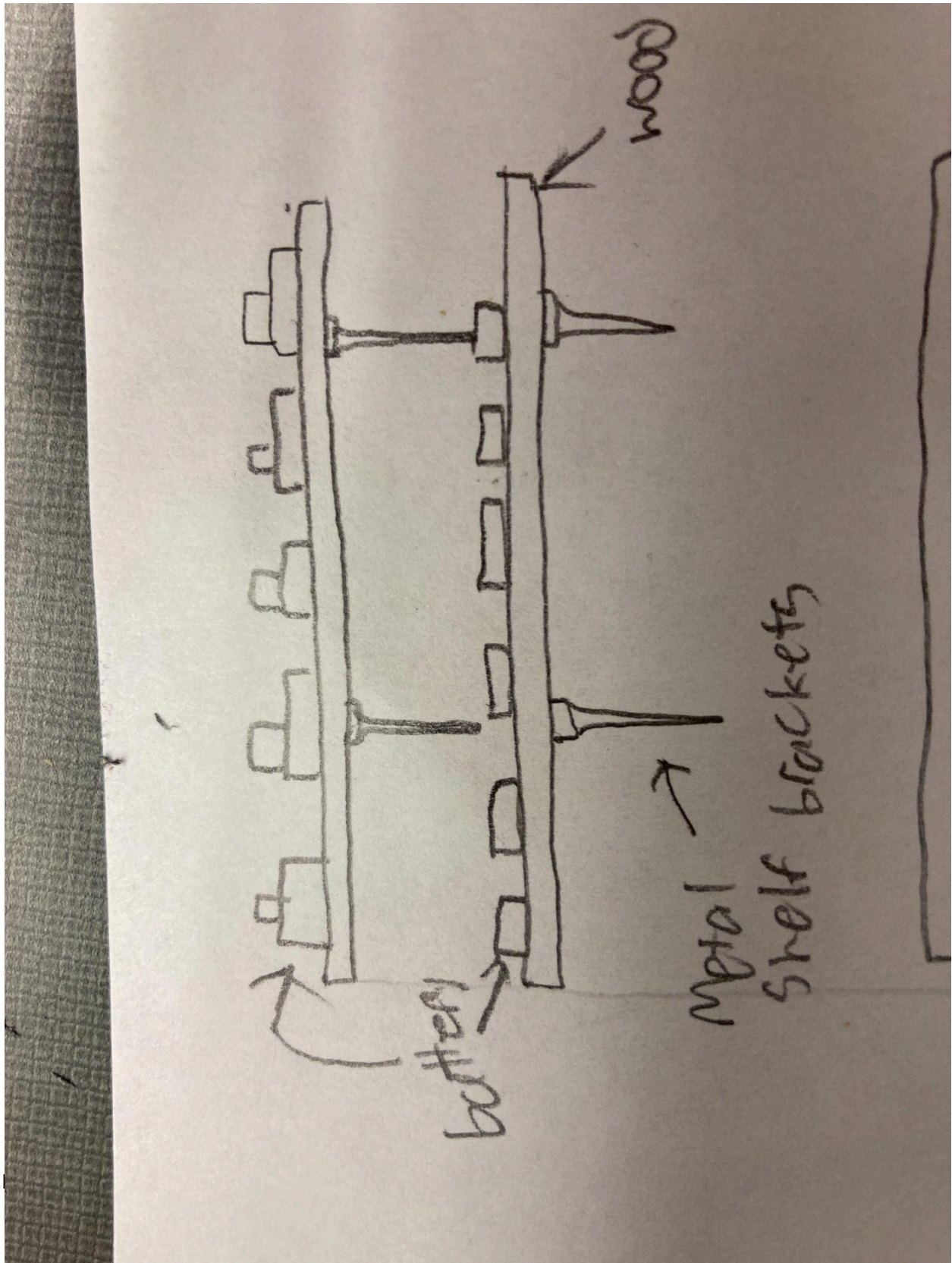
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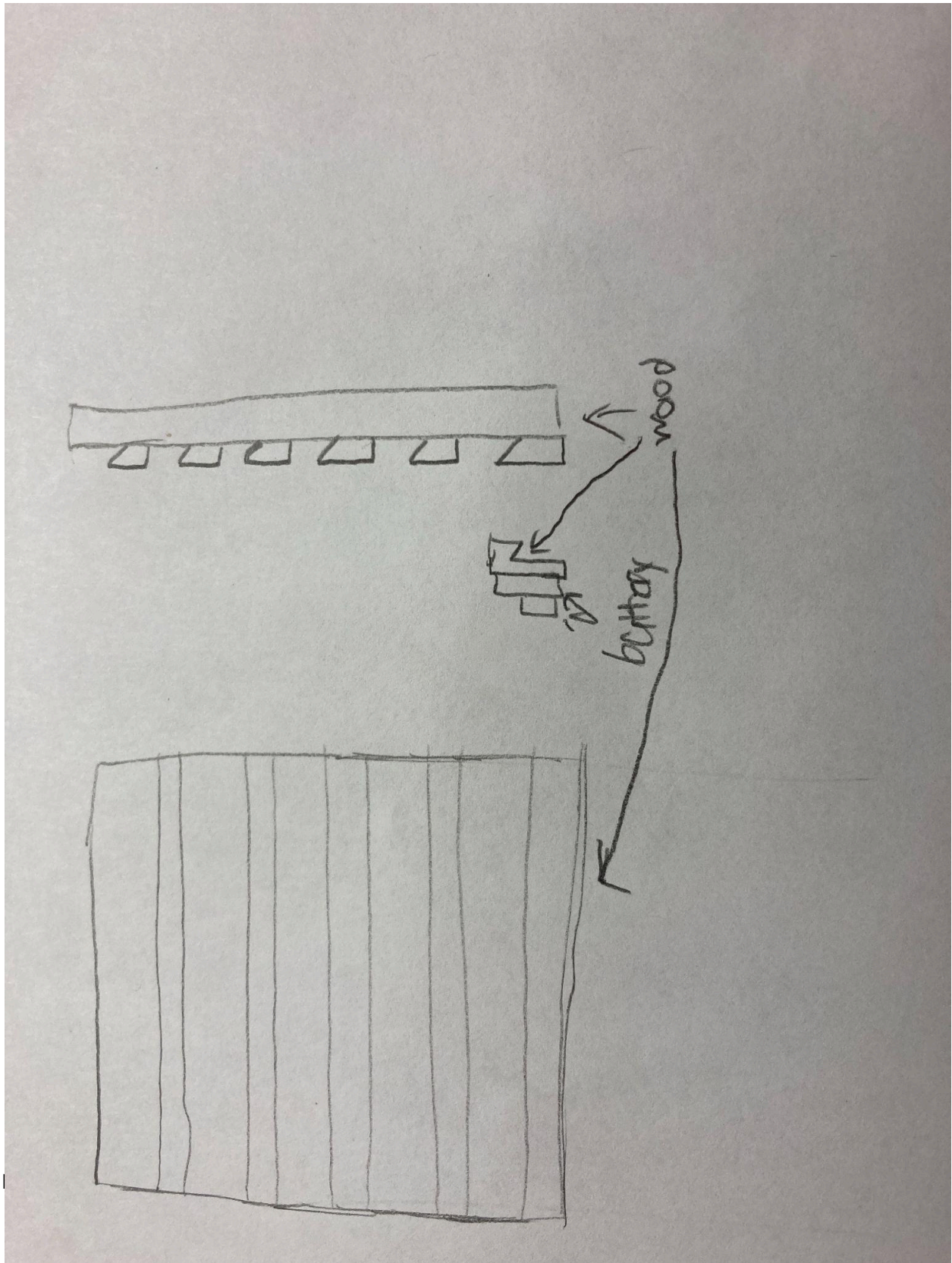
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needed.

Design Idea #1- Shelf

Date: 2/15/24

Description: shelves to hold battery chargers and extra batteries

Drawing that includes distinguishing information such as scale, dimensions, and materials:

Design Idea #2- Rack

Date: 2/15/24

Description: Battery chargers hang off of a rack on the wall with space for the cords to reach the outlet

Drawing that includes distinguishing information such as scale, dimensions, and materials:

Design Idea #3- peg board

Date: 2/15/24

Description: peg board with holders that the chargers and extra batteries would go into

Drawing that includes distinguishing information such as scale, dimensions, and materials:

Design Idea #4- French Cleat Wall

Date: 2/15/24

Description: cleats will hold the chargers on them and extra batteries. Will use hooks on the cleats to hold batteries

Design Idea #5- Drawer under table

Date: 2/15/24

Description: compact drawer under table. Two drawers stacked on top of each other; top drawer holds chargers, has a hole in the back for cords to go through. Bottom drawer holds extra batteries

Drawing that includes distinguishing information such as scale, dimensions, and materials:

Name: _____

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Drawing that includes distinguishing information such as scale, dimensions, and materials:

Design Idea Comparison and Selection (*D1, D2*)

When you've finished generating ideas, you must compare your design options according to all your design requirements. Incorporate the priority of those requirements when comparing design options. A decision matrix (perhaps a Pugh Matrix) is encouraged since it allows for systematic, weighted comparisons. If using weights in a decision matrix, be sure to explain and justify the weighting based on your design requirements. You may use the weighted design matrix template below, inserting your criteria and design options.

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<https://docs.google.com/spreadsheets/d/1EircGa8TwigacbPhYj9WpliddmSw0kRwHpewJFZX-Mw/edit?usp=sharing>

Design Blueprint (D4)

Describe all of your design's details such that someone unfamiliar with your project could build your prototype. Include sketches, dimensions, CAD (if taught in your class), manufacturing methods, material choices, supply lists, and costs.

Design Blueprint #1

Date:

Sketch:

Manufacturing Methods:

Step 1:

Step 2:

Step 3:

Supply List and Costs:

Item	Cost each	# needed	Total
Total Cost			

Plan of Action (D3)

A Gantt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.

